

DNS Threat Monitoring Using Splunk Enterprise

Executive Summary

This document describes the implementation of a DNS Threat Monitoring solution using Splunk Enterprise. DNS security logs were imported into Splunk, analyzed using SPL, visualized through dashboards, and investigated to identify abnormal DNS activity in a simulated SOC environment.

Project Overview

The project demonstrates DNS log ingestion, SPL query development, dashboard creation, report generation, alert use cases, and incident investigation using a CSV-based DNS dataset.

Lab Environment

- SIEM: Splunk Enterprise
- OS: Windows
- Dataset: dns_security_logs.csv
- Index: main
- Source Type: csv

Step 1 – Install Splunk Enterprise

Install Splunk Enterprise, access <http://localhost:8000>, and create the administrator account.

Step 2 – Prepare DNS Logs

Use a DNS CSV dataset containing Timestamp, Source_IP, Destination_IP, Query, Query_Type, Response_IP and Status fields.

Step 3 – Create Index

Settings → Indexes → New Index
Index Name: main

Step 4 – Upload DNS Logs

Settings → Add Data → Upload → dns_security_logs.csv

Source Type: csv

Index: main

Step 5 – Verify Data

SPL:

```
index=main source="dns_security_logs.csv"
```

Step 6 – DNS Traffic Analysis

Total DNS Requests :

```
index=main source="dns_security_logs.csv"  
| stats count
```

Top Queried Domains :

```
index=main source="dns_security_logs.csv"  
| top Query
```

Top Source Ips :

```
index=main source="dns_security_logs.csv"  
| top Source_IP
```

For Failed DNS Requests :

```
index=main source="dns_security_logs.csv"  
Status="NXDOMAIN"
```

Successful DNS Requests :

```
index=main source="dns_security_logs.csv"  
Status="Success"
```

Step 7 – Threat Detection

Excessive DNS Requests :

```
index=main source="dns_security_logs.csv"  
| stats count by Source_IP  
| where count>100
```

DNS Tunneling Detection :

```
index=main source="dns_security_logs.csv"  
| eval length=len(Query)  
| where length>50
```

Repeated Failed Queries :

```
index=main source="dns_security_logs.csv"  
Status="NXDOMAIN"  
| stats count by Source_IP  
| where count>20
```

Rare Domains :

```
index=main source="dns_security_logs.csv"  
| rare Query
```

Suspicious TLDs :

```
index=main source="dns_security_logs.csv"  
| search Query="*.xyz" OR Query="*.ru"
```

Most Queried Domains :

```
index=main source="dns_security_logs.csv"  
| chart count by Query
```

Step 8 – Dashboard Development

Dashboard: DNS Threat Monitoring Dashboard

Panels:

- Total DNS Requests (Single Value)
- Top Queried Domains (Bar Chart)
- Top Source IPs (Pie Chart)
- Failed DNS Queries (Table)
- Suspicious Domains (Table)
- DNS Traffic Over Time (Line Chart)

Timeline Query:

```
index=main source="dns_security_logs.csv"  
| timechart count
```

Step 9 – Alert Use Cases

Alert 1 – Excessive DNS Requests

```
index=main source="dns_security_logs.csv"  
| stats count by Source_IP  
| where count>100
```

Alert 2 – NXDOMAIN Spike

```
index=main source="dns_security_logs.csv" Status="NXDOMAIN"  
| stats count  
| where count>50
```

Alert 3 – Suspicious Domains

```
index=main source="dns_security_logs.csv"  
| search Query="*.xyz"
```

Step 10 – Incident Investigation

Scenario: A host generated approximately 500 DNS requests.

Investigation Questions:

- Which Source IP generated the requests?
- Which domains were queried?
- Were the DNS requests successful?
- Was DNS tunneling attempted?
- Which endpoint was affected?
- What is the event timeline?

Representative Queries:

```
index=main source="dns_security_logs.csv"  
| stats count by Source_IP  
| sort -count
```

```
index=main source="dns_security_logs.csv"  
| top Query limit=20
```

```
index=main source="dns_security_logs.csv"  
| stats count by Status
```

```
index=main source="dns_security_logs.csv" Source_IP="192.168.1.50"  
| eval Length=len(Query)
```

```
| table Query Length
```

```
| sort -Length
```

```
index=main source="dns_security_logs.csv"
```

```
| stats count by Source_IP
```

```
| sort -count
```

```
index=main source="dns_security_logs.csv"
```

```
| timechart count
```

Findings

The project successfully demonstrated DNS log ingestion, SPL-based analysis, dashboard creation, report generation, and incident investigation within Splunk Enterprise.

Conclusion

This implementation provides hands-on experience with SIEM operations, DNS monitoring, SPL development, and SOC investigation workflows using simulated DNS log data.